

STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	1-3 Semester (2025-26)
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Branch	CSE
Course Outcome	5

Case Study Topic: IPL Ad-Impression Archive Sorting using External Sorting and K-Way Merge

Work Done Summary:

In this case study, an IPL streaming platform generated approximately **50 GB of timestamped ad-impression logs** during the 2024 IPL season. Since the available RAM is only **8 GB**, the complete dataset cannot be sorted entirely in memory.

To solve this problem, the **External Merge Sort** technique was implemented. The dataset was divided into multiple memory-sized chunks, each chunk was sorted independently in RAM, and the sorted chunks were written back to disk. Finally, a **K-Way Merge Algorithm** using a Min-Heap was applied to merge all sorted chunks into one globally sorted output file.

Implementation Details:

1. Chunk creation from large input data.
2. In-memory sorting of chunks.
3. K-Way Merge using Min Heap.
4. Disk-based output generation.
5. I/O cost analysis.
6. Comparison with Binary Merge.

Result:

Number of Chunks :

6

Merge Technique Used :

K-Way Merge

Data Structure Used :

Min Heap (Priority Queue)

Final Sorted Output :

8
10
11
12
15
19
22
25
28
30
33
38
41
47
49
51
55
60
75
80
85
88
89
92

External Merge Sort successfully sorted a large IPL ad-impression archive that exceeded available memory. By dividing the data into chunks, sorting them individually, and merging them using a K-Way Min Heap Merge, the system produced a globally sorted output efficiently. This approach is widely used in large-scale database systems, search engines, and big-data processing platforms.

Output:

Merged Output:

```
8 10 11 12 15 19 22 25 28 30 33 38 41 47 49 51 55 60 75 80 85 88 89 92
```

Merged Output:

```
8 10 11 12 15 19 22 25 28 30 33 38 41 47 49 51 55 60 75 80 85 88 89 92
```

GitHub Link:

<https://github.com/saisharan134/DSA2-Case-Studies>

Student Signature	Faculty Signature
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